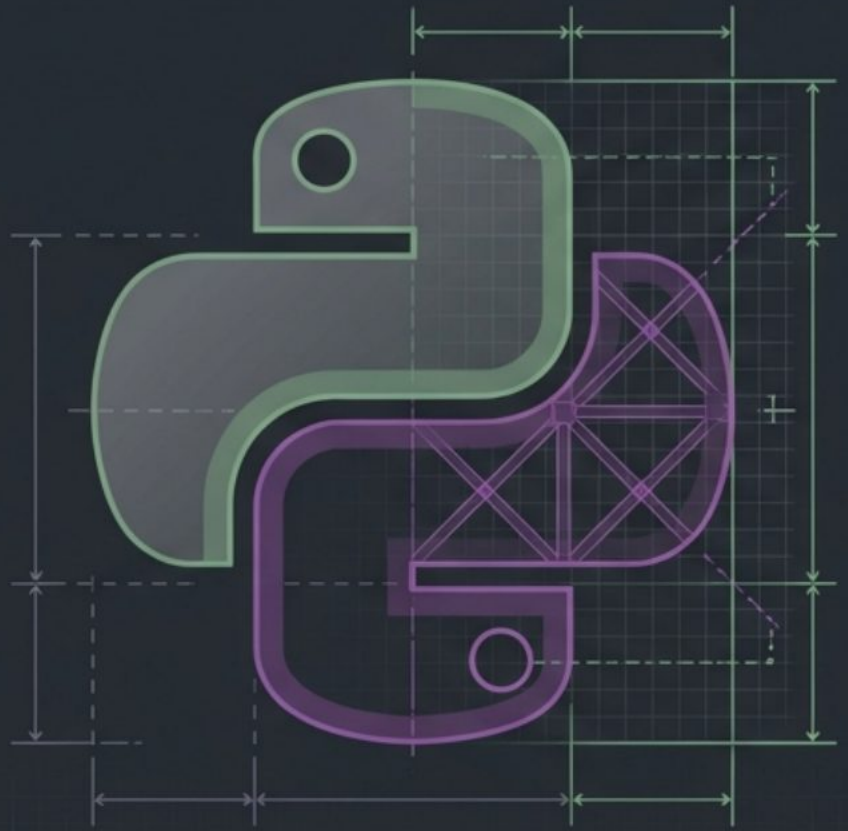
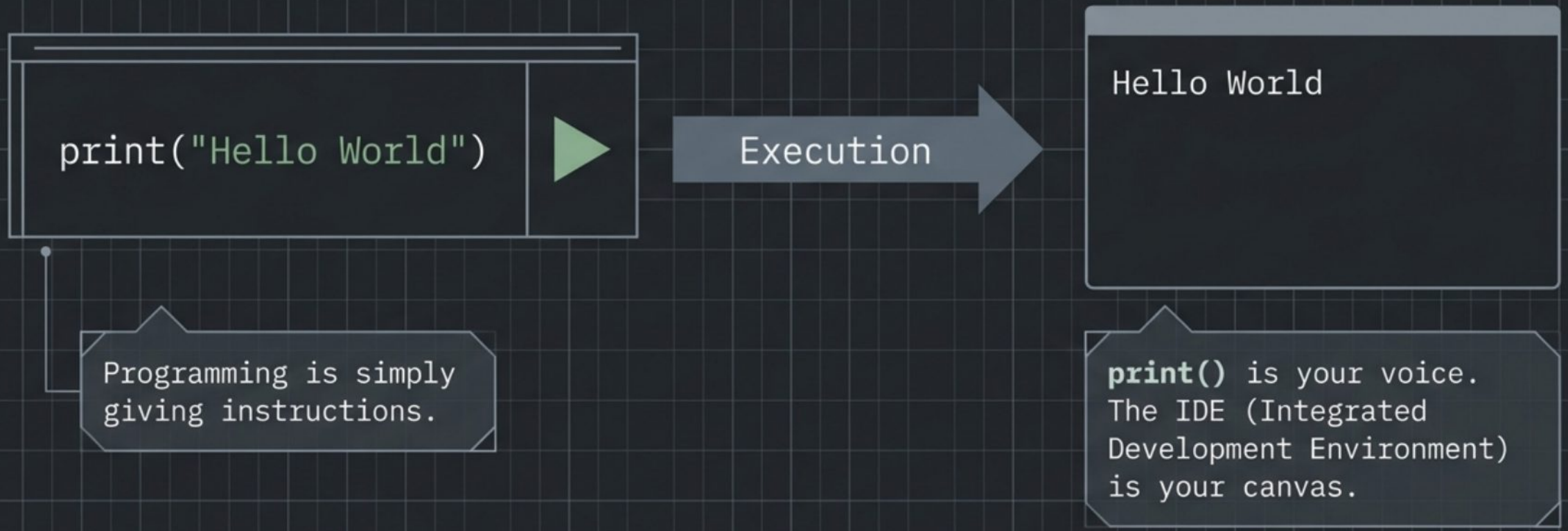
The Python Blueprint

From First Script to First AI: A 30-Minute Architecture Course



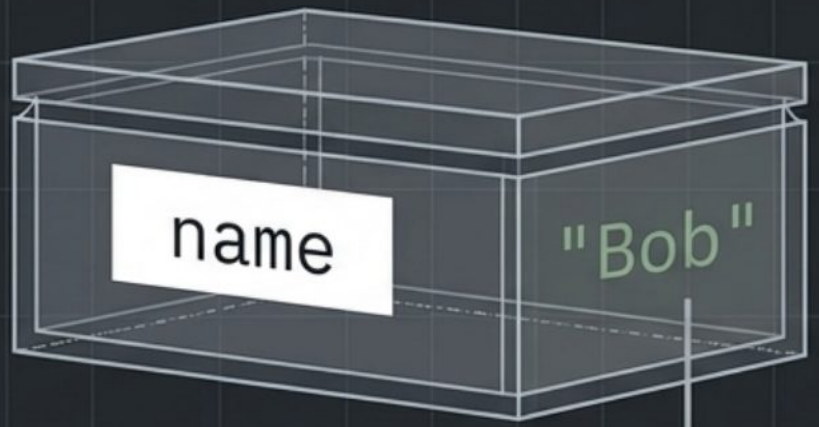
Gemini Notebook

Speaking the Machine's Language



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Variables: Naming Your Memory



Variables store data so you can reuse and update it dynamically.

```
print("Hello " + name + "!")
```



Rigid, older method prone to errors.

```
print(f"Hello {name}!")
```

F-Strings (`f"..."`) allow you to seamlessly drop variables directly into your text using curly brackets {}.

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The Raw Materials (Atomic Data Types)

Python needs to know the exact material you are working with. You cannot do math with a String, just as you wouldn't build a foundational wall out of glass.

String (Text)



"Apple"
"Bob"

Integer (Whole Numbers)



10
-10

Float (Decimals)



10.5, 3.1415

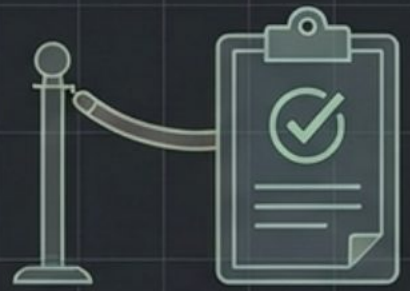
Boolean (Binary State)



True, False

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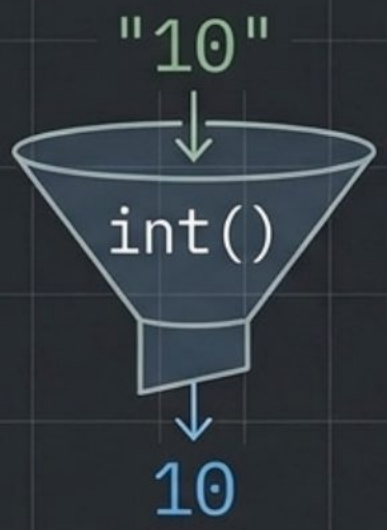
Structuring Complex Data

List `[]`	Tuple `()`	Set `{}`	Dictionary {k:v}`
			
Mutable, Ordered	Immutable, Ordered	Mutable, No Duplicates	Key-Value pairs

`names = ['Agnetha', 'Björn', 'Benny', 'Anni-Frid']`	`coordinates = (2.5, 1.5)`	`numbers = {1, 2, 3, 3, 4, 4, 5}`
		`print(numbers)`
		`# Output: {1, 2, 3, 4, 5}`
			`scores = {'Bob': 1, 'James': 2}`

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Enforcing the Rules (Type Safety)



Converting messy scraped text into usable math.



Annotations don’t change how code runs, but they act as digital guardrails, warning the developer if they try to put text inside a number variable.

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The Logic Gates: Making Comparisons

Before a program can make a decision, it must weigh the conditions. Every complex comparison ultimately reduces to a single binary state: **True** or **False**.

`1 > 2`



`1 < 2`



`2 ≥ 2`



`1 ≤ 2`



`1 == 1` (Is exactly equal to)

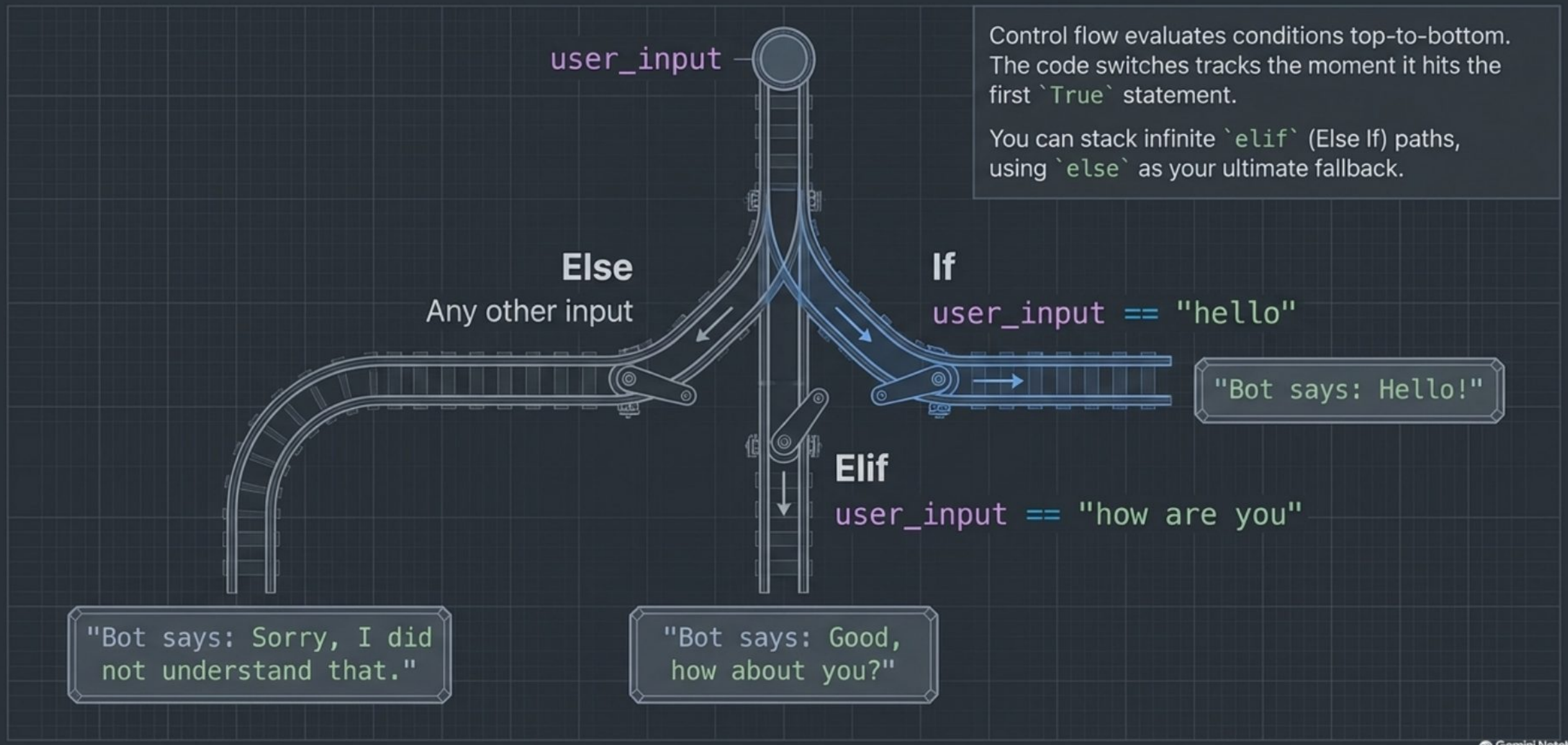


`1 != 2` (Is NOT equal to)



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Forking the Path (Control Flow)



Control flow evaluates conditions top-to-bottom. The code switches tracks the moment it hits the first `True` statement.

You can stack infinite `elif` (Else If) paths, using `else` as your ultimate fallback.

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The Engines of Automation: For vs. While

For Loop




```
for i in range(3):
    print("hello")
```

Finite iteration: Runs exactly as many times as the items in a sequence (or list).

While Loop

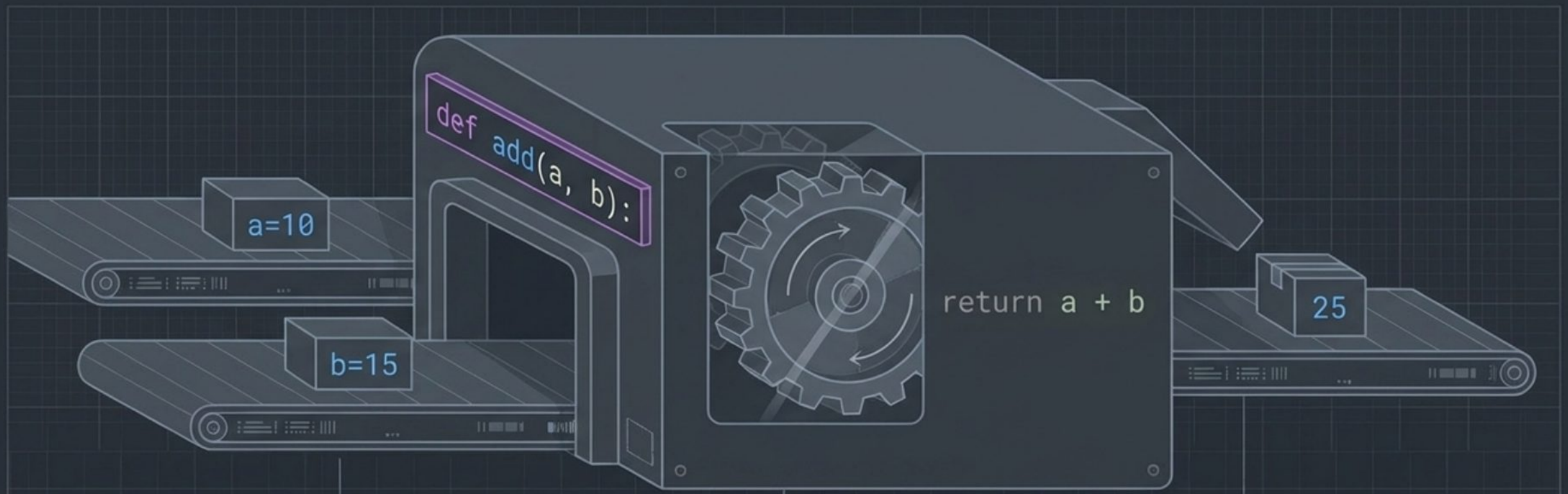


```
while True:
    print("hello")
```

Infinite iteration: Runs forever until a specific condition forces the loop to break or evaluate to ‘False’.

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Functions: Reusable Logic Factories



Stop repeating yourself. Functions encapsulate logic into a reusable machine.

You can set default behaviors directly on the intake (e.g., `def greet(name, greeting='Hi')`). If a greeting isn't provided, the factory uses its default.

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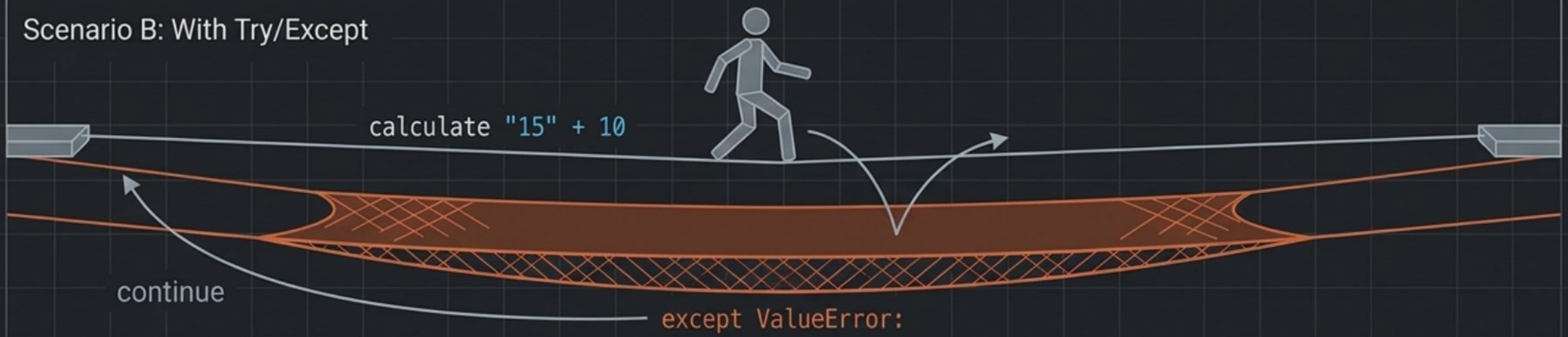
The Safety Net (Try / Except)

Users will break your app. Professional developers anticipate this. Wrapping dangerous code in a `try` block allows you to catch specific errors and ask the user to fix their input, rather than letting the entire application fatally crash.

Scenario A: Without Try/Except

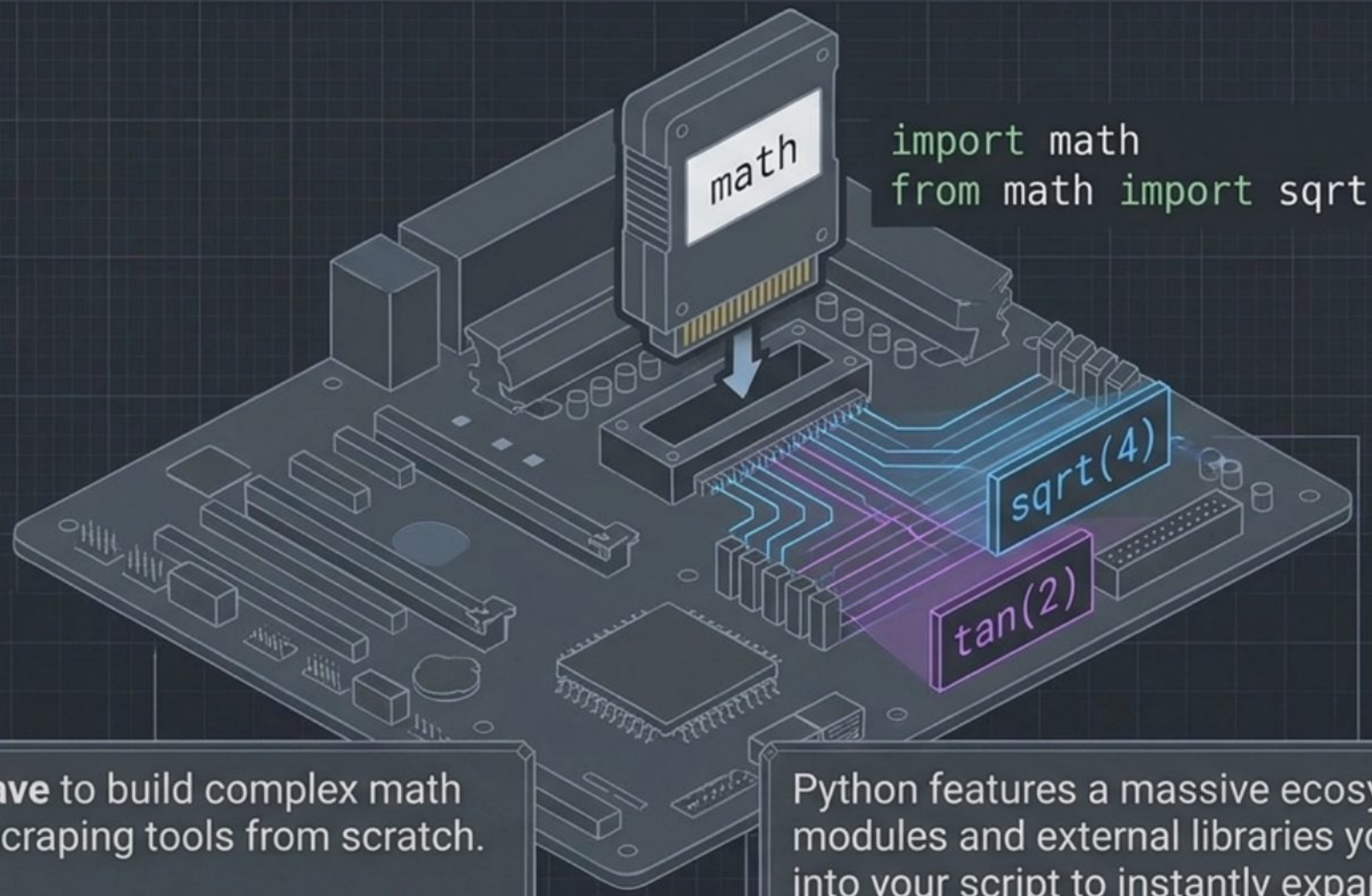


Scenario B: With Try/Except



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Borrowing Superpowers (External Imports)

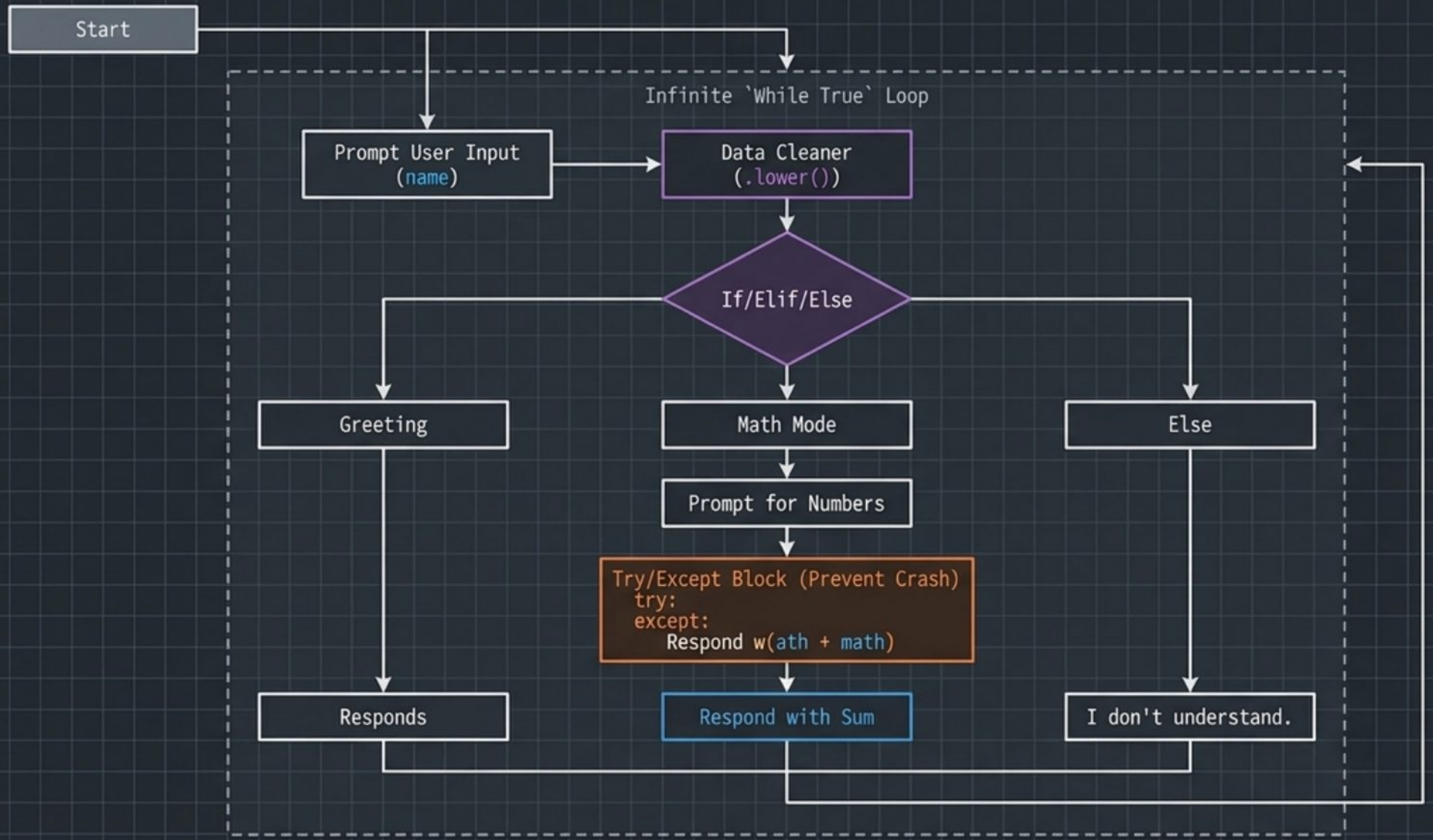


You don't have to build complex math or internet-scraping tools from scratch.

Python features a massive ecosystem of built-in modules and external libraries you can plug directly into your script to instantly expand its capabilities.

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Synthesis Phase I: Chatbot Architecture



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Synthesis Phase II: The Finished Machine

1



The infinite loop engine keeping the bot alive.

2



Standardizing data so 'Hi' and 'hi' are treated equally.

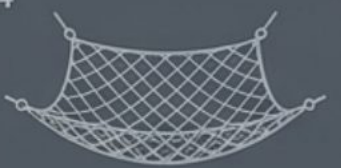
```
while True:
    user_input = input("You: ").lower()
    if user_input in ['hi', 'hello']:
        print("Bot: Hi there!")
    elif user_input in ['add', '+']:
        try:
            num1 = float(input("First number: "))
            num2 = float(input("Second number: "))
            print(f"Bot: {num1 + num2}")
        except ValueError:
            print("Bot: Invalid number.")
    else:
        print("Bot: I don't understand.")
```

3



Control flow switching to math mode.

4



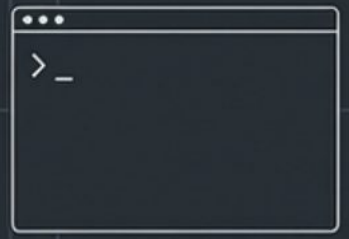
Catching letters when expecting numbers to prevent a crash.

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The Developer's Mindset







3 Build Real Apps

The only way to truly learn is to pick a project you care about, encounter errors, and solve them.

2 Google Everything

Even developers with 5 years of experience Google syntax constantly. It is part of the job.

1 Start Small

Don't try to build Facebook on day one. Master the basics first.

You now have the fundamental blueprint. Welcome to Python.

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